

Problem 1

List1 in case(a) remains the same as [1,2,3,4] since there is nothing in the script to change it while in case(b) it changes to [1,2,3,4,161] since list2 is connected to list1 when list2 is modified list1 is modified as well.

Problem 2

```
# Aidan, Noah, Zach, Luis
# AerE 1610, Lab 2, Problem 2
# Write a script that generates a list of random integers each in the range (-10,50).
Perform
# each of the following and print all results to the screen
# a) Find the number of negative elements in the list.
# b) Find the sum of all positive elements in the list.
# c) Find the smallest positive element in the list.
# d) Find the second smallest positive element in your list.
import random

# generate list
nums = [random.randint(-10, 50) for _ in range(15)]
print("List:", nums)

# a) number of negatives
neg_count = sum(1 for x in nums if x < 0)
print("Negative count:", neg_count)

# b) sum of positives
pos_sum = sum(x for x in nums if x > 0)
print("Sum of positives:", pos_sum)

# c) smallest positive
positives = [x for x in nums if x > 0]
if positives:
    smallest_pos = min(positives)
    print("Smallest positive:", smallest_pos)
else:
    print("No positive numbers")

# d) second smallest positive
```

```
if len(positives) >= 2:
    positives_sorted = sorted(positives)
    second_smallest = positives_sorted[1]
    print("Second smallest positive:", second_smallest)
else:
    print("Not enough positive numbers")
```

Output :

List: [-3, 4, 28, -7, 26, 35, 2, 39, 40, 5, 30, 45, 30, 42, -3]

Negative count: 3

Sum of positives: 326

Smallest positive: 2

Second smallest positive: 4

Problem 3

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AerE 1610H, Lab 2, Problem 3

Write a script that generates a list of 10 random integers in the range (1,20).

Rearrange the elements into new list the following way: all even numbers first, then all odd numbers.

```
import random
```

```
# we use random.randint to generate random integers 1-20
```

```
numbers = [random.randint(1,20) for x in range(10)]
```

```
print("Original list: ", numbers)
```

```
#this is how we classify them into evens or odds
```

```
even = [x for x in numbers if x % 2 == 0]
```

```
odds = [x for x in numbers if x % 2 != 0]
```

```
updated_list = even + odds # here we add the odds to the evens
```

```
print("Updated list: ", updated_list)
```

Output :

Original list: [2, 8, 17, 12, 5, 18, 16, 4, 19, 17]

Updated list: [2, 8, 12, 18, 16, 4, 17, 5, 19, 17]

Problem 4

Aidan, Noah, Zach, Luis
 # AerE 1610, Lab 2, Problem 2
 #Write a script that will generate random integers in the range from 0 to 50, and
 #print them, until one is finally generated that is greater than 25. The script should
 #print how many attempts it took.

```
import numpy as np #import the numpy
```

```
num = 0 # set number to 0
```

```
attempts = 0 # set number of attempts to 0
```

```
while num <= 25: # while number is less than 25, continue looping
```

```
    num = np.random.randint(0, 51) # generate a random number between 0 and 51
```

```
    attempts += 1 # add to attempts
```

```
    print(num) # print each number
```

```
print("Attempts:", attempts) # print the amount of attempts taken
```

Output :

14

16

30

Attempts: 3

0

17

3

20

28

Attempts: 5